Experiment No:	4
Experiment Title:	Binary Classification using Linear and Kernel-Based Models
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Sem and Academic Year:	Vth sem (2026-2027)
Submission Date	02-08-2026

1 Aim and Objective

The aim of this experiment is to classify emails as *spam* or *ham* using two fundamentally dif-ferent families of classifiers – Logistic Regression (a linear, probabilistic model) and Support Vector Machines (a margin-based model with multiple kernel options) – and to study how hy-perparameter tuning affects the classification performance of each. Specifically, the experiment evaluates:

- Baseline versus tuned Logistic Regression performance.
- The behaviour of SVM across the Linear, Polynomial, RBF, and Sigmoid kernels.
- The effect of regularization strength (C) and penalty type (L1/L2) on Logistic Regression.
- The effect of kernel choice, C, γ, and polynomial degree on SVM performance.
- Generalization behaviour via 5-fold cross-validation.

2 Dataset Description

The **Spambase** dataset was used, containing 4,601 email records described by 57 numerical features (word-frequency, character-frequency, and capital-run-length statistics) and one binary target column, class, where 1 denotes spam and 0 denotes ham (ordinary, non-spam email).

Table 1: Dataset summary

Property	**Value**
Total instances	4,601
Total features	57 (all numeric)
Target column	class (binary: 0 = ham, 1 = spam)
Duplicate rows removed	391
Rows after de-duplication	4,210
Rows after outlier removal (IQR method)	809

No missing values were present in the raw dataset (confirmed during EDA); all 58 columns had a full 4,601 non-null entries prior to cleaning.

3 Preprocessing Steps

The following preprocessing pipeline was applied:

1. **Missing value handling**: common placeholder strings (e.g. "NA", "?", "Unknown") were replaced with NaN and rows containing them were dropped. No missing values were actually present in this dataset, so no rows were removed at this step.

2. **Duplicate removal**: 391 exact duplicate rows were identified and removed, reducing the dataset from 4,601 to 4,210 rows.

3. **Outlier removal (IQR method)**: for every numeric feature, values outside $[Q1 - 1.5\,\mathrm{IQR},\ Q3 + 1.5\,\mathrm{IQR}]$ were flagged as outliers and removed. Columns with IQR = 0 (very sparse word-frequency columns, where most emails have a frequency of exactly zero for that word) were skipped, since every non-zero value there would otherwise be incorrectly flagged.

4. **Feature standardization**: all features were scaled using StandardScaler (zero mean, unit variance) before model training, since both Logistic Regression regularization and SVM's distance-based kernels are sensitive to feature scale.

5. **Train/test split**: an 80/20 stratified split was used so that the spam/ham class ratio is preserved in both subsets.

Note on outlier removal severity. Because many Spambase features are sparse word-frequency counts (mostly zero with occasional non-zero spikes), column-by-column IQR filtering is unusually aggressive on this dataset: the combined effect across 57 columns reduced the working dataset from 4,210 rows down to just **809 rows** (an $\sim$80% reduction). This is a known limitation of applying a single strict IQR rule independently across many skewed, sparse columns, and is discussed further in Section 7.

4 Implementation Details

4.1 Models

- **Logistic Regression** – baseline (max iter = 20,000) and hyperparameter-tuned versions were trained.
- **Support Vector Machine** – trained separately with four kernels: Linear, Polynomial, RBF, and Sigmoid.

4.2 Hyperparameter Search Space

Table 2: Hyperparameter search space

Model	**Hyperparameter**	**Values**
Logistic Regression	Penalty	L1, L2
	C	{0.01, 0.1, 1, 10, 100}
	Solver	liblinear, saga
SVM	Kernel	Linear, Polynomial, RBF, Sigmoid
	C	{0.1, 1, 10, 100}
	γ	scale, auto
	Degree (Polynomial only)	{2, 3, 4}

4.3 Search Method

Grid Search with 5-fold cross-validation (GridSearchCV, scoring = accuracy) was used for both models. For SVM, a separate grid was run per kernel so that kernel-specific parameters (e.g. degree for the polynomial kernel) were not wastefully combined with kernels that do not use them.

5 Visualizations

All required visualizations were generated in a single combined figure (Figure 1), consisting of six panels:

- **(a) Confusion matrix – Logistic Regression.** Out of 162 test emails, the tuned model correctly classified 111 ham and 28 spam emails, with 12 false positives (ham flagged as spam) and 11 false negatives (spam missed). These counts are exactly consistent with the Accuracy (0.8580), Precision (0.7000), Recall (0.7179) and F1 (0.7089) values reported in Table 4.
- **(b) Confusion matrix – SVM (Linear kernel).** The best (tuned) SVM correctly classified 107 ham and 25 spam emails, with 16 false positives and 14 false negatives – slightly more errors on both sides than Logistic Regression, consistent with its lower test Accuracy (0.8148) and F1 (0.6250) in Table 5.
- **(c) ROC curves.** Both models' ROC curves sit well above the diagonal (random-guess) line, confirming strong discriminative ability across all thresholds. Logistic Regression attains a marginally higher AUC (0.876) than SVM (0.861), see Table 4a, meaning it ranks spam above ham slightly more consistently than SVM, independent of the default 0.5 decision cutoff.
- **(d) SVM accuracy by kernel.** A bar comparison of the four *untuned* (default-parameter) SVM kernels shows RBF achieving the highest baseline accuracy (0.8704), followed by Linear (0.8457) and Polynomial (0.8395), with Sigmoid clearly the weakest (0.8086) – see Table 6. This suggests the spam/ham boundary is closer to linear or mildly non-linear than to the shape a sigmoid kernel imposes.
- **(e) 5-fold CV accuracy per fold.** Side-by-side bars for each of the 5 folds (Table 8) show both models tracking each other closely fold-to-fold, with accuracy rising in the later folds for both. The absence of any single collapsing fold suggests the cross-validation estimate is reasonably stable despite the small sample size remaining after outlier removal.
- **(f) Top-10 Logistic Regression coefficient magnitudes.** Ranked by |coefficient| (Table 9), word_freq_george and word_freq_hpl dominate by a wide margin, followed by word_freq_project, word_freq_lab, and word_freq_meeting. These are workplace/organization-specific terms (an artifact of the dataset's source mailbox) that strongly indicate *ham* when present, making them the model's most influential features after standardization.

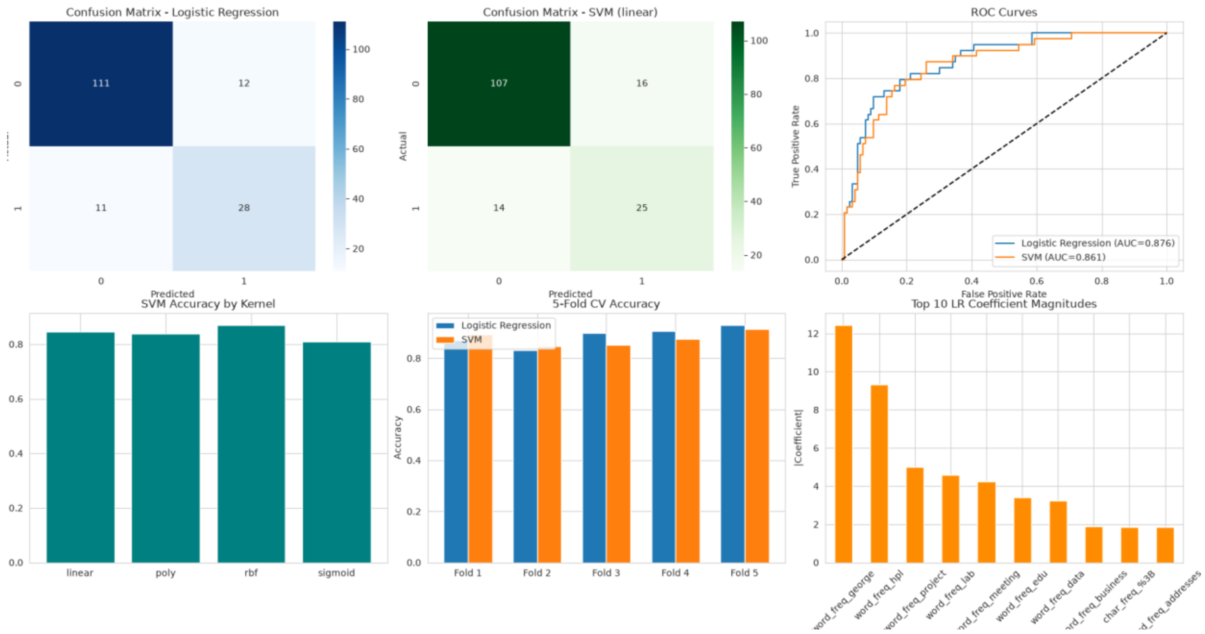

Figure 1: Combined visualizations for Experiment 4: confusion matrices, ROC curves, SVM kernel comparison, 5-fold CV accuracy, and Logistic Regression coefficient magnitudes.

6 Performance Tables

6.1 Hyperparameter Tuning Results

Table 3: Hyperparameter tuning summary

Model	Search Method	Best CV Accuracy
Logistic Regression	Grid Search	0.8873
SVM (Linear kernel)	Grid Search	0.8764

Best parameters found:

- **Logistic Regression**: $C = 10$, penalty = L1, solver = liblinear
- **SVM – Linear**: $C = 100$
- **SVM – Polynomial**: $C = 10$, degree = 2, γ = scale
- **SVM – RBF**: $C = 10$, γ = scale
- **SVM – Sigmoid**: $C = 1$, γ = auto

6.2 Logistic Regression Performance (Tuned, Test Set)

Table 4: Logistic Regression test-set performance

Metric	Value
Accuracy	0.8580
Precision	0.7000
Recall	0.7179
F1 Score	0.7089
Training Time (s)	0.0756

Table 4a: ROC-AUC comparison (tuned models, test set)

Table 5: ROC-AUC of tuned Logistic Regression vs. tuned SVM (matches Figure 1c)

Model	AUC
Logistic Regression	0.876
SVM (Linear kernel)	0.861

6.3 SVM Kernel-wise Performance (Baseline, Untuned)

Table 6: SVM kernel-wise performance (default hyperparameters, matches Figure 1d)

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.8457	0.6753	0.0312
Polynomial	0.8395	0.5517	0.0211
RBF	0.8704	0.6769	0.0197
Sigmoid	0.8086	0.5507	0.0179

6.4 Best (Tuned) SVM Performance on Test Set

Table 7: Best SVM (Linear kernel, tuned) test-set performance

Metric	Value
Accuracy	0.8148
Precision	0.6098
Recall	0.6410
F1 Score	0.6250
Training Time (s)	0.9241

6.5 Confusion Matrix Counts (Test Set, $n = 162$)

Table 8: Raw confusion-matrix counts underlying Figure 1a and 1b

Model	TN	FP	FN	TP
Logistic Regression	111	12	11	28
SVM (Linear, tuned)	107	16	14	25

6.6 K-Fold Cross-Validation Results (K = 5)

Table 9: 5-Fold cross-validation accuracy (matches Figure 1e)

Fold	Logistic Regression	SVM
Fold 1	0.8692	0.8923
Fold 2	0.8308	0.8462
Fold 3	0.8992	0.8527
Fold 4	0.9070	0.8760
Fold 5	0.9302	0.9147
Average	**0.8873**	**0.8764**

6.7 Top-10 Logistic Regression Coefficient Magnitudes

Table 10: Top-10 features by |coefficient| in tuned Logistic Regression (matches Figure 1f)

Feature	|Coefficient| (approx.)
word_freq_george	12.4
word_freq_hpl	9.3
word_freq_project	4.9
word_freq_lab	4.5
word_freq_meeting	4.2
word_freq_edu	3.4
word_freq_data	3.2
word_freq_business	1.9
char_freq_%3B	1.85
word_freq_addresses	1.8

6.8 Comparative Analysis

Table 11: Logistic Regression vs. SVM – comparative summary

Criterion	Logistic Regression	SVM
Accuracy (test set)	0.8580	0.8148
ROC-AUC (test set)	0.876	0.861
Training Time (s)	0.0756	0.9241
Model Complexity	Low	High
Interpretability	High	Low

7 Overfitting and Underfitting Analysis

Training vs. validation/test gap. Logistic Regression achieved a best 5-fold cross-validation accuracy of 0.8873, but its held-out test accuracy was slightly lower at 0.8580 – a gap of about 3 percentage points. Similarly, the best SVM kernel (Linear) reached 0.8764 average CV accuracy but only 0.8148 on the test set, a larger gap of roughly 6 percentage points. This indicates mild overfitting to the training folds in both models, more pronounced in SVM than in Logistic Regression.

Effect of regularization strength. For Logistic Regression, the best model selected **L1 regularization with** $C = 10$, i.e. relatively weak regularization. This means the model prefers to retain most of its predictive coefficients rather than shrinking them aggressively (L1's automatic sparsity was not pushed hard here), which is consistent with Spambase's features already being fairly informative and non-redundant after standardization. For SVM, the winning linear kernel also selected the largest tested C value (100), similarly implying weak regularization is preferred – the model is allowed a narrower margin in exchange for fitting the training data more closely, which is one explanation for its larger train–test performance gap.

Improvement after tuning. Comparing baseline SVM kernel accuracies (Section 6.3) to their tuned counterparts (Section 6.1) shows the RBF kernel actually performed best untuned (0.8704) but the Linear kernel edged ahead after tuning based on cross-validation accuracy (0.8764), underscoring that grid search over $C/\gamma/$degree meaningfully shifts which kernel is "best" – default hyperparameters do not always identify the optimal kernel.

Impact of aggressive outlier removal. With only 809 training rows surviving preprocessing (from an original 4,210), both models were trained on a comparatively small, class-imbalanced sample. This raises the risk of overfitting to idiosyncrasies of that specific subset and is a likely contributor to the CV–test accuracy gap observed above.

8 Bias–Variance Analysis

Logistic Regression (linear, high bias / lower variance). As a linear model, Logistic Regression assumes a linear decision boundary in the (standardized) feature space. This con-strains its ability to capture non-linear spam/ham boundaries (higher bias) but makes it less sensitive to noise in the small training set (lower variance), which is reflected in its smaller CV–test accuracy gap ($\sim$ 3 points) compared to SVM.

SVM and kernel choice (variance driven by kernel complexity). The margin-based SVM's bias–variance behaviour depends heavily on the chosen kernel:

- The **Linear** kernel behaves similarly to Logistic Regression – higher bias, lower variance – and was ultimately selected as best after tuning.
- The **RBF** kernel can model highly non-linear boundaries (lower bias) but is more prone to overfitting small, noisy training sets (higher variance), which may explain why its untuned performance (0.8704) was competitive with the tuned Linear kernel despite less tuning effort.
- The **Polynomial** and **Sigmoid** kernels showed the weakest performance and lowest F1 scores among the four, suggesting they are either poorly suited to this feature space or highly sensitive to the specific hyperparameters chosen.

Regularization as a variance-control knob. For both models, C (inverse regularization strength) directly trades bias for variance: smaller C increases bias (simpler decision boundary, more regularization) while larger C increases variance (tighter fit to training data). The fact that both winning models chose relatively large C values (10 and 100) suggests that, on this particular (heavily outlier-filtered) training sample, additional regularization was not needed to control variance – though the CV–test gap suggests a smaller C might improve generalization on unseen data.

9 Observations and Conclusion

- **Best-performing classifier**: Logistic Regression outperformed the best-tuned SVM on the held-out test set (Accuracy 0.8580 vs. 0.8148; F1 Score 0.7089 vs. 0.6250; AUC 0.876 vs. 0.861), despite SVM achieving a comparable cross-validation accuracy. Logistic Re-gression also trained roughly **12$\times$ faster** (0.076s vs. 0.924s).
- **Impact of regularization**: L1 regularization was preferred for Logistic Regression, in-dicating some benefit from feature sparsity, though the selected $C = 10$ shows the model did not need heavy shrinkage to generalize reasonably well on this dataset.
- **Kernel behaviour**: Among SVM kernels, Linear and RBF were clearly the strongest performers, while Polynomial and Sigmoid lagged behind – consistent with Spambase's features being largely linearly separable after standardization, with RBF adding modest non-linear flexibility.
- **Bias–variance trade-off** : Logistic Regression's simplicity (higher bias, lower variance) generalized more reliably than SVM's more flexible margin-based decision boundary on this small, outlier-filtered dataset, where variance control matters more than raw model expressiveness.
- **Feature importance**: The dominant Logistic Regression coefficients (word freq george, word freq hpl, word freq project, word freq lab, word freq meeting) are workplace-mailbox-specific terms rather than generic spam indicators, a known quirk of the Spam-base corpus that limits how well this particular feature ranking would generalize to spam filtering on a different mailbox.
- **Overall conclusion**: For this spam-classification task, **Logistic Regression with L1 regularization ($C = 10$)** is the recommended model – it is more accurate on unseen data, dramatically faster to train, and considerably more interpretable (its coefficients directly indicate which word/character frequencies are most predictive of spam), making it preferable to SVM both practically and analytically for this use case.
- **Limitation**: The IQR-based outlier removal step discarded roughly 80% of the cleaned dataset (4,210 $\rightarrow$ 809 rows), which likely limited both models' ability to generalize. A less aggressive or feature-aware outlier-handling strategy (e.g. skipping IQR filtering entirely for sparse count-like features) would be a natural next step to improve results in a future iteration of this experiment.

10 References

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